



SIMATS ENGINEERING

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Course Code: CSA1205

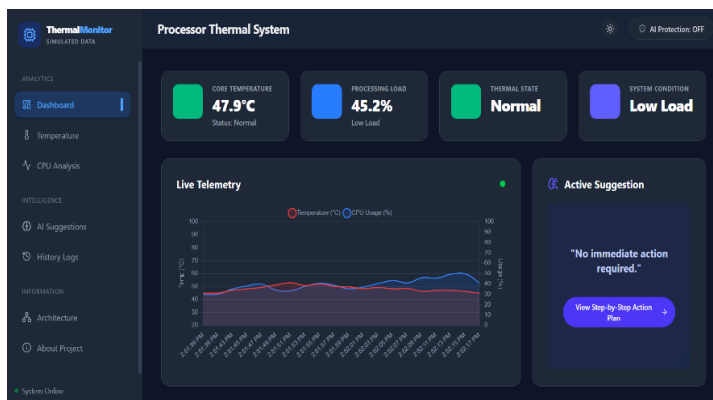
Slot: C

Course Name: Computer Architecture

Course Faculty: Dr .K. Vijaya Bhaskar

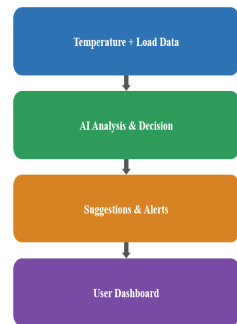
Project Title: Processor Thermal Monitoring System with Real-Time CPU Analysis and AI-Based Suggestions

Module Photographs:



Module 2: AI Suggestion Module

- This module combines the temperature readings and the analysed CPU load into a single thermal state.
- Based on that state it generates practical, prioritised recommendations for the user.
- Typical suggestions include closing heavy background applications, improving airflow or cleaning dust from the vents.
- When a critical threshold is crossed, an immediate notification is raised on the React dashboard.
- Every suggestion and alert is logged with a timestamp so that thermal behaviour can be reviewed later.



Project Description: Real-Time Thermal Monitoring and AI Suggestion Evaluation

The Real-Time Thermal Monitoring and AI Suggestion module forms the core of the Processor Thermal Monitoring System with Real-Time CPU Analysis and AI-Based Suggestions. The module continuously tracks CPU core temperatures, clock speeds, and load percentages using system sensor data, presenting live readings through an interactive dashboard. The system analyzes temperature trends over time to detect abnormal heating patterns, potential thermal throttling, and cooling inefficiencies before they impact performance. An AI-based recommendation engine processes the collected data to suggest corrective actions, such as adjusting fan speed, reducing background workload, or improving airflow, based on the severity of the thermal condition.. Performance is measured using factors such as monitoring accuracy, response time to thermal spikes, and the practical usefulness of the AI-generated suggestions. This module ensures that the developed system helps users maintain optimal processor health, extend hardware lifespan, and prevent performance degradation caused by overheating.

Student Signature

Guide Signature